Maastricht University

Master Thesis Research Proposal

Multi-agent Reinforcement Learning for the Collaborative Vehicle Routing Problem

Name: André-Ignace Ghonda Lukoki

ID: i6330047

Programme: MSc Econometrics and Operations Research

Supervisor: Uğur Güler, Frank Phillipson, Rui Jorge Almeida

Date: January 22, 2026

School of Business and Economics

Master

© Academic year 2025-2026 Maastricht University School of Business and Economics

1 Motivation and Conceptual Framework

1.1 Research Aim and Motivation

Collaboration in the form of shared resources and routing between firms or players in the transportation industry has received greater interest in recent years (Guajardo & Rönnqvist, 2016) as it is said to result in lower transportation costs and greater efficiency thanks to shorter travel distances, better vehicle utilization and route planning (Mak, Xu, Pearce, Ostroumov, & Brintrup, 2023), reduced greenhouse gas emissions, congestion, and noise pollution (Gansterer & Hartl, 2018), increased flexibility against demand fluctuations and disruptions (Guajardo & Rönnqvist, 2016) such as vehicle or driver shortages, and the ability to reach greater economies of scale and greater profit.

Collaboration has also received some interest in the industry, one example is the creation of System Alliance Europe, which defines itself as a pan-European logistics alliance, in 2005 (SAE, 2026). However, larger industry adoption requires the creation of fair cost allocation mechanisms that can scale to a large number of participants as collaborating firms can only hope to obtain the benefits outlined above if they can reach an agreement regarding the split of the collaboration gains. Additionally, those collaborations can only be sustained over time if all participating firms are convinced that the gain-sharing agreement is implemented fairly.

1.2 Conceptual Framework

The vehicle routing problem (VRP) is a generalization of the traveling salesman problem (TSP) that seeks to determine a set of non-overlapping tours for a group of m vehicles (Gansterer & Hartl, 2018). In the capacitated VRP (CVRP), each vehicle is associated with a maximal capacity q_i and visits a set a customers (nodes) with demand d_j such that the total demand on each route does not exceed the capacity of the vehicle on that route. In this thesis, the collaborative VRP, an extension to the capacitated VRP, involves vehicles operated by different firms aiming to serve their customers at minimal cost. The problem links a cost allocation problem to the CVRP and seeks to determine a suitable mechanism to split the gains (i.e. the cost savings) obtained from collaboration alongside a set of suitable routes (Guajardo & Rönnqvist, 2016).

Cooperation in the transport industry can take multiple forms, a centralized collaboration effort involves a central authority with complete information organizing the cooperation efforts while decentralized settings contain firms that might not desire to share much information leading to central

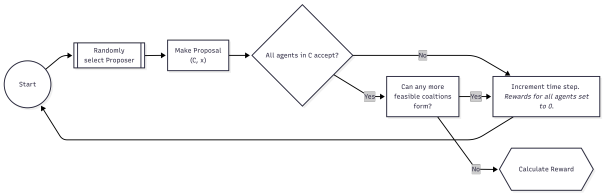

Figure 1: Flowchart of the Coalitional Bargaining game used by Mak et al. (2023).

authorities with limited information and information asymmetries among the participants (Gansterer & Hartl, 2018). Additionally, cooperation between firms operating at the same level in the market is termed horizontal collaboration (Luo, Zhou, & Lev, 2022). This work considers collaboration as a decentralized effort between firms competing in the same market.

In line with Mak et al. (2023), the collaborative VRP is modeled as a coalitional bargaining game (N, v) with transferable utility (see Figure 1) where N is the set of players and v is the characteristic function that assigns a value $v(S) \in \mathbb{R}$ to any coalition $S \in 2^N$. The aim of the coalitional bargaining game is to derive a payoff distribution $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n_+$ that is efficient, *meaning that* $\sum_{i \in N} x_i = v(N)$, and stable, *i.e. it satisfies coalitional rationality, meaning that* $\sum_{i \in S} x_i \geq v(S)$ *for all nonempty coalitions* $S \in 2^N$. That is, any solution must lie in the core $C(N, v)$ defined as the set of all efficient and stable payoff distributions and this set can contain zero, one, or infinitely many payoff distributions (Peters, 2015).

According to Guajardo and Rönnqvist (2016) and Gansterer and Hartl (2018), the Shapley value and the Nucleolus are two solution concepts that are often proposed in the literature on collaborative vehicle routing. Some relevant properties of both concepts include uniqueness and efficiency, however the Shapley value does not always satisfy coalitional rationality (Peters, 2015).

An additional consideration in this problem relates to fairness. In the industry, long-lasting partnerships can only be sustained if all parties believe that their contributions are adequately reflected in the allocation of gains. However, asymmetries in bargaining power often allow some firms to appropriate a share of the surplus that exceeds their relative contribution to the alliance. As a result, a fair allocation may need to account not only for agents' contributions, but also for their relative bargaining power represented by their ability to form valuable coalitions. Each of the proposed solution concepts can be interpreted as emphasizing one of those dimensions. The Shapley value allocates costs based on the average marginal contribution of the agents and the nucleolus minimizes the 'dissatisfaction', or excess defined as $e(S, \mathbf{x}) = v(S) - \sum_{i \in S} x_i$, of each coalition (Guajardo &

Rönnqvist, 2016).

Coalitional bargaining games are well-equipped to model the dynamics in the road transport industry due to the fragmentation of the industry (Mak et al., 2023). For this reason, it is reasonable to model collaboration initiatives in the transportation industry as a decentralized bargaining process between rational, self-interested firms competing in the same market. However, coalitional bargaining games typically model static environments assuming that the bargaining occurs after the characteristic function is known. By contrast, stochastic games, which are an extension of Markov decision processes to multiple agents and often used in multi-agent reinforcement learning (MARL), describe an environment in which multiple players repeatedly interact to influence both their rewards and the state of the environment (Albrecht, Christianos, & Schäfer, 2024).

Stochastic games are defined by a finite set of agents N, a set of states S including terminal states $\bar{S} \subset S$ and an initial state s_0, a finite set of actions A_i and a reward function $R_i : S \times A \times S \to \mathbb{R}$ for each agent $i \in N$, a state transition probability function $\mathcal{T} : S \times A \to S$, and a discount factor $\gamma \in [0, 1]$ influencing the interest afforded to future rewards by the agents (Mak et al., 2023).

Modeling the bargaining process as a stochastic game also reflects important practical realities. In this setting, the allocation problem shifts from merely computing coalitional values to learning them through repeated interactions with other players (Mak et al., 2023). In the industry, firms, seeking to maintain a degree of operational privacy, may refuse to fully disclose their internal information. This means that cooperation must develop gradually, as repeated interactions allow agents to build trust, exchange information, and adapt their strategies over time. Thus, by modeling the bargaining process as a stochastic game, multi-agent reinforcement learning naturally emerges as a mechanism to model these decentralized interactions while approximating fair and stable collaborative outcomes.

Lastly, the application of MARL to other domains involving self-interested or cooperative agents such as (video) games (Mak et al., 2023), autonomous routing, and traffic scheduling (Ning & Xie, 2024) suggests potential for similar outcomes in collaborative routing problems. In line with the recommendations of Guajardo and Rönnqvist (2016), this thesis investigates approximate methods to determine suitable gain-sharing agreements.

1.3 Research Question

This thesis aims to investigate the following research question:

To what extent can MARL approximate a fair cost allocation among self-interested

agents in a collaborative vehicle routing problem?

To address this question, we seek to determine whether multi-agent reinforcement learning can aid in determining an efficient set of routes and a fair allocation of the collaboration gains among a set of self-interested, cost-minimizing agents in a collaborative vehicle routing setting. While the primary focus of this research is evaluating the fairness of the resulting cost allocations, efficient route planning is a necessary component in determining the collaboration gains to be used in the allocation process.

Thus, the objectives of this research are threefold:

1. To build upon the MARL framework for the collaborative vehicle routing problem proposed by Mak et al. (2023) to generate cost allocation proposals.

2. To implement cooperative game-theoretic solution concepts, namely the Shapley value and the nucleolus to investigate whether MARL approximates allocations that are consistent with these fairness criteria.

3. To evaluate the scalability and performance of the MARL framework on instances involving up to four agents. The evaluation considers solution quality (measured by deviations from the nucleolus), solution fairness (measured by correlation with Shapley value), and computational efficiency (measured by time to convergence), and solution value (measured by comparing allocations to the baseline bots in Mak et al. (2023)).

1.4 Added Value

In their survey, Gansterer and Hartl (2018) observe that much of the prior literature on the collaborative vehicle routing problem has treated the transportation or the cost sharing aspect of the problem in isolation. Little attention has been paid to the development of approaches that consider the two simultaneously. This thesis aims to address this gap by proposing the use of multi-agent reinforcement learning to jointly learn efficient routing policies and approximate gain-sharing agreements in a collaborative setting.

Many traditional approaches assume explicit knowledge of the characteristic function or rely on the computation of a series of linear programs, both of which do not always scale well with the number of agents (Mak et al., 2023). Instead, this thesis investigates whether multi-agent reinforcement learning can provide a scalable alternative to approximate payoff distributions. By considering a

set of self-interested decision-makers evolving in a shared routing environment, MARL enables the emergence of routing strategies and payoff distributions without requiring the complete enumeration of the characteristic function (Mak et al., 2023).

By evaluating instances involving up to four agents, this work assesses the performance and stability of the MARL framework as the cooperative environment grows and is expected to provide new insights into the computational tractability of the collaborative VRP. Lastly, this thesis also obtains insights on the trade-offs between routing efficiency, fairness, and stability in collaborative routing problems.

2 Intended Methods

2.1 Computational Methods

In this study, the collaborative vehicle routing problem involves a set of self-interested agents aiming to serve a subset of customers at minimal costs where agents may collaborate by sharing customers, therefore reducing total costs. The environment is modeled as a routing problem with known customer locations and demands in which collaboration gains may arise from joint route planning. These gains must be allocated among the participating agents.

To address the research question, the study draws on the MARL framework developed by Mak et al. (2023). Each agent is modeled as an autonomous decision-maker (decentralized execution) with the ability to propose a coalition with an associated cost allocation or to accept or reject a coalition proposal. Centralized training with decentralized execution (CTDE) is employed as it allows agents to learn cooperative behavior while retaining individual objectives. The environment, state representations, and training procedures proposed by Mak et al. (2023) serve as a baseline for investigation into alternative techniques in the MARL literature including topics on transfer learning, skills retention, and generalization from n to $n+1$ agents in hopes to develop a scalable MARL framework.

Due to their widespread use in the literature on collaborative vehicle routing (Gansterer & Hartl, 2018), cooperative game-theoretic solution concepts such as the nucleolus and the Shapley value are selected as fairness criteria. Those solution concepts are computed using exact methods defined below for each instance and the value of a coalition is defined as the minimum routing cost achieved by that coalition. For instances involving a small set of agents, a brute-force approach to compute

the coalitional values should be computationally tractable (Mak et al., 2023).

The methods to compute the solution concepts are informed by the literature. The nucleolus is computed using a series of linear programs using the technique proposed by Guajardo and Jörnsten (2015) which ensures the correctness in the implementation and avoids the common computational pitfalls highlighted in their study. On the other hand, the Shapely value is computed by means of the following formula presented in Guajardo and Rönnqvist (2016),

$$x_j = \sum_{S \subseteq N : j \in S} [\frac{(n - |S|)!(|S| - 1)!}{n!}][v(S) - v(S \backslash \{j\})], \forall j \in N$$

Consistency of the allocation proposals with the aforementioned solution concepts is investigated by means of correlation analysis, similarly to the work of Mak et al. (2023).

The experimental design focuses on instances involving two to four agents which has the benefit of enabling the computation of exact game-theoretic benchmarks while allowing the comparison of solution quality, fairness and value for a range of environment sizes. For each configuration (2, 3, or 4 agents), multiple instances of the collaborative VRP are generated by varying customer locations and demand. The coalition structure and the cost allocation proposal are recorded for each instance in the training set. The experiments are repeated with multiple random seeds in line with (Mak et al., 2023).

The performance of the MARL framework is evaluated along the four dimensions outlined in the research objectives. Solution quality is measured by the absolute and relative deviations from the nucleolus for each agent. The nucleolus is chosen as a measure of solution quality due to the guarantee that its associated allocation always lies in the core, whenever it is non-empty (Guajardo & Rönnqvist, 2016). Solution fairness is assessed using the correlation between the MARL allocations and the implied by the Shapley value. This comparison captures the extent to which the proposed allocations align with the marginal contributions of the agents. Computational efficiency is evaluated by means of the time to convergence, defined as the number of time steps required to reach a state in which no additional coalition can form. This metric assesses whether agents learn to propose more acceptable allocations and thus reach stable agreements faster. Lastly, the definition of baseline bots to evaluate the solution value aims to determine whether the setting is not too simplistic and ensure that agents are learning interesting behavior (Mak et al., 2023).

In conclusion, this methodology seeks to provide insights into the extent to which multi-agent

reinforcement learning can provide fair cost allocations in the collaborative vehicle routing problem and whether such an approach offers a scalable alternative to the current game-theoretic methods.

3 Sustainable Development Goals (SDGs)

Transportation accounts for a significant portion of global greenhouse gas emissions, estimated at around 15% in 2023 (Fernández, 2024) and according to EEA (2025), road transportation was responsible for 73% of those emissions in the European Union. This has lead the European Union to adopt emission reduction targets in 2030 and 2050 affecting both commercial and personal road transportation (EC, n.d.). Collaborative vehicle routing has the potential to reduce GHG emissions in the transportation industry, as a smaller fleet or shorter travel distances are often achieved through successful horizontal collaboration. This directly aligns with the 11th SDG as collaboration has the potential to reduce the transportation industry's involvement in congestion and pollution in and around cities.

4 Time Plan

The timeline spanning from February to July aims to reflect the practical constraints of this master thesis project. The plan includes monthly discussion meetings with all supervisors while allowing for the inclusion of ad hoc meetings to address questions and issues arising from the thesis work progression with the relevant supervisor. Additionally, the sentence 'discussion with supervisors' highlight activities that might require increased interactions with supervisors.

Period 4

When	What
February	Targeted literature review on MARL including topics on parameter sharing, transfer learning, skills retention and n to $n+1$ generalization. Definition of MARL environment design, state representations, and training techniques (discussion with supervisors). Setup Python and cloud environment (discussion with supervisors). Finalize problem and evaluation metric formulation.
March	Implement instance generator for up to four agents inspired by Mak et al. (2023). Implement exact Shapley value computation and baseline bots. Implement an algorithm to compute coalition values. Reproduce elements of the MARL framework of Mak et al. (2023) (discussion with supervisors). Validate MARL environment, instance generation, and baseline bots (discussion with supervisors). Thesis: write literature review.
April	Implement exact nucleolus computation. Validate Shapley and nucleolus computation on generated instances. Development of MARL coalition proposal mechanism (discussion with supervisors).

Period 5

When	What
April	MARL training (2-3) agents (discussion with supervisors). Validation of MARL framework on small instances (discussion with supervisors). Evaluations (quality, fairness, and value). Thesis: write methods section (start).
May	MARL training (3-4) agents. Multi-seeds experimental runs (discussion with supervisors). Visualize results. Thesis: write methods section (continued).
June	Evaluations (quality, fairness, and value, comp. efficiency). These: write results and experimental setup (discussion with supervisors).

Period 6

When	What
June	Thesis: write discussion, conclusion, and introduction. Revisions, feedback and thesis defense preparation.
July - Sept	Thesis defense rehearsals. Thesis submission. Thesis Defense.

5 References

References

Albrecht, S. V., Christianos, F., & Schäfer, L. (2024). *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT Press. Retrieved from https://www.marl-book.com

EC. (n.d.). *European Climate Law*. Retrieved from https://climate.ec.europa.eu/eu-action/european-climate-law_en

EEA. (2025). *Greenhouse gas emissions from transport in Europe*. Retrieved from https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-transport

Fernández, L. (2024). *Distribution of greenhouse gas emissions worldwide in 2023, by sector*. Retrieved from https://www.statista.com/statistics/241756/proportion-of-energy-in-global-greenhouse-gas-emissions/

Gansterer, M., & Hartl, R. F. (2018). Collaborative vehicle routing: A survey. *European Journal of Operational Research*, *268*(1), 1–12. Retrieved from https://www.sciencedirect.com/science/article/pii/S0377221717309360 doi: https://doi.org/10.1016/j.ejor.2017.10.023

Guajardo, M., & Jörnsten, K. (2015). Common mistakes in computing the nucleolus. *European Journal of Operational Research*, *241*(3), 931–935. Retrieved from https://www.sciencedirect.com/science/article/pii/S0377221714008595 doi: https://doi.org/10.1016/j.ejor.2014.10.037

Guajardo, M., & Rönnqvist, M. (2016). A review on cost allocation methods in collaborative transportation. *International Transactions in Operational Research*, *23*(3), 371–392. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1111/itor.12205 (_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/itor.12205) doi: https://doi.org/10.1111/itor.12205

Luo, C., Zhou, X., & Lev, B. (2022). Core, shapley value, nucleolus and nash bargaining solution: A Survey of recent developments and applications in operations management. *Omega*, *110*, 102638. Retrieved from https://www.sciencedirect.com/science/article/pii/S0305048322000469 doi: https://doi.org/10.1016/j.omega.2022.102638

Mak, S., Xu, L., Pearce, T., Ostroumov, M., & Brintrup, A. (2023). Fair collaborative vehicle routing:

A deep multi-agent reinforcement learning approach. *Transportation Research Part C: Emerging Technologies*, *157*, 104376. Retrieved from https://www.sciencedirect.com/science/article/pii/S0968090X23003662 doi: https://doi.org/10.1016/j.trc.2023.104376

Ning, Z., & Xie, L. (2024). A survey on multi-agent reinforcement learning and its application. *Journal of Automation and Intelligence*, *3*(2), 73–91. Retrieved from https://www.sciencedirect.com/science/article/pii/S2949855424000042 doi: https://doi.org/10.1016/j.jai.2024.02.003

Peters, H. (2015). *Game Theory* (2nd ed.). Springer Berlin, Heidelberg. Retrieved from https://link.springer.com/book/10.1007/978-3-662-46950-7 doi: https://doi.org/10.1007/978-3-662-46950-7

SAE. (2026). *System Alliance Europe.* Retrieved from https://www.system-alliance.eu/